

CSE465 — Project Progress Report

Replacing Moshi’s Helium-7B Backbone with Qwen2.5-0.5B on a Single T4 GPU

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Project Overview

Moshi is a speech-to-speech AI model by Kyutai. Its text backbone, Helium-7B, needs about 14 GB of GPU memory, making it impossible to fine-tune on free cloud GPUs (Tesla T4, 15.6 GB). We replaced Helium-7B with the much smaller Qwen2.5-0.5B (500M parameters), using LoRA adapters and 4-bit quantization, so the full system fits on one T4 GPU. All other parts of Moshi (Mimi audio codec, depth decoder, generation harness) were kept unchanged.

Work Completed

Architecture: HybridBackbone

Qwen2.5-0.5B uses 896-dim hidden states; Moshi expects 4096. We built a **HybridBackbone** module with two trainable linear layers to bridge the gap, without changing Moshi’s parent class.

- **Input projector** Win: 4096 → 896 dim (≈3.67M parameters)
- **Output projector** Wout: 896 → 4096 dim (≈3.67M parameters)
- **LoRA adapters** (rank 16, alpha 32) on all attention and MLP layers inside Qwen (≈8.80M parameters)

Total trainable: ≈150M / 1.35B total (11.16%). Everything else is frozen.

Memory Optimizations

Peak training VRAM measured at **2.98 GB** via `torch.cuda.memory_allocated()`.

- 4-bit NF4 quantization (double-quant) on both Moshi and Qwen via BitsAndBytes
- Mimi cast to bfloat16 after loading (saves ≈140 MB)
- Gradient checkpointing on Qwen and the depth decoder
- 8-bit AdamW optimizer — cuts optimizer state memory by ~4×
- Training clips capped at 2 seconds (25 frames at 12.5 Hz)

Bug Fixes in Moshi’s Released Code

Two bugs were found and fixed that completely blocked the pipeline on T4:

- **Mimi padding integer overflow:** Padding was read as $\approx 9.2 \times 10^{18}$ on T4 hardware, crashing convolutions. Fixed by converting padding to native Python integers.
- **Depth decoder dtype mismatch:** Depth decoder loaded in float16 while the rest used bfloat16, causing a matrix multiplication error. Fixed by matching dtypes at load time.

Dataset and Training Configuration

Dataset: **DailyTalkContiguous** (Kyutai) — 200 dialogues from 2,541 available, split 90/10 (180 train, 20 validation). Stereo 24 kHz audio with aligned transcripts.

Learning rate:	2×10^{-4}	Total steps:	1,000
LR schedule:	warmup (50 steps) + cosine	Batch size:	4 (gradient accum. 4)
Optimizer:	AdamW8bit	Grad clipping:	norm 1.0

Resource Comparison

Metric	Original (Helium-7B)	Modified (Qwen2.5-0.5B)	Change
Parameters	7 Billion	0.5 Billion	−93%
VRAM (peak)	~14 GB	~2.98 GB	−79%
T4 Compatible	No	Yes	—
Disk Space	~15 GB	~1 GB	−93%

Training Results (From Notebook Logs)

The training loop ran successfully for 1,000 steps and survived a Colab session restart at step 700, resuming correctly at step 900 with no data loss. However, the model did not learn — validation loss went up, not down.

Step	Loss	Perplexity	Accuracy	F1	BLEU
200	7.59	1,972	0.0104	0.0017	0.52
300	8.30	4,015	0.0042	0.0010	0.52
400	8.72	6,136	0.0031	0.0007	0.45
500	8.76	6,354	0.0010	0.0004	0.46
600	9.31	11,013	0.0031	0.0031	0.46
700	9.56	14,189	0.0042	0.0029	0.44
900 [†]	7.72	2,257	0.0031	0.0008	0.57
1000	7.79	2,423	0.0031	0.0008	0.58

[†] Colab restarted at step 700; optimizer state was reset. Not real improvement.

Key findings from the log:

- Loss rose from 7.59 (step 200) to 9.56 (step 700) — the model was getting worse, not better.
- Token accuracy and F1 stayed below 0.02 at every checkpoint throughout training.
- Audio codebook accuracy was exactly **0.0** at every checkpoint — no correct audio tokens were ever predicted.
- End-to-end generation produced a 4.96-second waveform from a 3-second prompt. Listening confirmed it was **pure noise**. Whisper-tiny transcription returned silence or random words.

Why Training Failed: Root Cause Analysis

Three specific structural problems caused the failure:

1. **Padding tokens were included in the loss.** The loss function was called without `ignore_index`, so $\approx 70\%$ of each label sequence (padding positions 20–63 of 64) was included in training. The model was pulled in two directions at once — predict real words and predict the pad token — so the gradients never pointed in a useful direction. *Fix: set `ignore_index=tokenizer.pad_token_id`.*
2. **The input projector randomly scrambled Moshi’s input embedding.** Moshi’s parent class adds the text embedding and all audio embeddings together and passes the combined 4096-dim result directly to the decoder. Our input projector Win was randomly initialized, so it mapped this meaningful input to a random direction in 896-dim space at every step. Qwen received meaningless input throughout all 1,000 training steps. LoRA adapters of rank 16 do not have enough capacity to fix this.
3. **The frozen depth decoder received inputs it had never seen.** Moshi’s depth decoder was trained together with Helium-7B and expects its specific output distribution. Our output projector Wout was randomly initialized, so the depth decoder received random conditioning at every step. The text-token loss alone gives no signal about what the depth decoder needs, so it could never produce correct audio tokens. This explains the audio accuracy of exactly 0.0 throughout.

Next Steps

The failure is specific to this setup. It does not mean backbone replacement is impossible. Planned fixes:

1. **Mask padding in the loss** — add `ignore_index` to remove the conflicting gradient signal.
2. **Add a distillation loss** — run Helium-7B on the same input and train Wout to match its output via mean squared error. This gives a dense learning signal at every position.
3. **Unfreeze the depth decoder** — add LoRA adapters so it can learn to read the new backbone’s output distribution.
4. **Use a larger student model** — Qwen2.5-1.5B or 3B would reduce the $14\times$ capacity gap while still fitting on a paid-tier GPU.
5. **More data and longer training** — the full 2,500-dialogue DailyTalk corpus for at least 50,000 steps is a more realistic minimum.